

PARTS LIST

Semiconductors:

IC1	- MT8870 DTMF decoder
IC2	- 7805, 5V voltage regulator
D1	- 1N4007, rectifier diode
T1	- BC547 npn transistor
T2	- 2N2222 npn transistor
Board1	- Arduino Uno board

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R2	- 2.2-kilo-ohm
R3	- 560-ohm
R4	- 2.7-kilo-ohm
R5, R6	- 100-kilo-ohm
R7	- 300-kilo-ohm
R8	- 1-kilo-ohm

Capacitors:

C1	- 1uF, 35V electrolytic
C2	- 4.7uF, 35V electrolytic
C3	- 100uF, 35V electrolytic
C4, C5	- 0.1uF ceramic

Miscellaneous:

CON1	- 2-pin connector for audio jack
RL1	- 5V, single-changeover relay
M1	- 12V, 10rpm geared DC motor
BATT.1, BATT.2	- 12V battery
XTAL1	- 3.58MHz
	- 3.5mm stereo jack
	- Male-to-female jumper wires
	- 7-pin male connector for Arduino
	- 2-pin terminal connector for batteries
	- Heatsink for 7805

0V (ground) for low.

digitalRead(). This reads the value from a specified digital pin, as either high or low.

begin(). This sets the data rate in bits per second (baud) for serial data transmission. For communicating with the computer, use one of these rates: 300, 600, 1200, 2400, 4800, 9600, 14400, 19200, 28800, 38400, 57600 or 115200. You can also specify other rates. For example, to communicate with an external component over pins 0 and 1 of Arduino, a particular baud rate may be required.

Construction and testing

An actual-size PCB layout of the food dispenser is shown in Fig. 4 and its components layout in Fig. 5. Use a geared DC motor for rotating the lid. A 12V battery (BATT.1) is used to power the transistor amplifier stage, 7805 and Arduino board. Another 12V battery (BATT.2) is used to power the geared DC motor.

Connect the DTMF module and receiver phone using a 3.5mm audio



EFY Note

The source code of this project is available for free download at source.efymag.com

jack. Make sure that the receiver phone is on auto-answering mode. Now, make a call on the receiver phone. After the call is received, press 2 on the caller phone. If communication is established successfully, RL1 will be energised and motor M1 will start rotating its shaft. The lid will rotate along with the shaft. For stopping the motor, press 5 on the caller phone.

Note. You may use a readymade DTMF module in place of IC1, and a single-changeover relay module in place of RL1 and D1 used here. **EFY**



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